IMO Training – Number Theory
Level 2 Assignment 1

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Instructions:

- Write your solutions legibly on separate sheets of paper.
- Each question carries 7 marks.
- Complete solutions with justified arguments are required for full credit in every question.
- Both paper and electronic copies are accepted.
- Deadline for submission: 10 August, 2019

1. Find all intergers m, n such that $m^5 - n^5 = 16mn$.
2. Find all positive integers m, n such that $125 \cdot 2^n - 3^m = 271$.
3. Prove that there exist infinitely many positive integers n such that $\dfrac{4^n + 2^n + 1}{n^2 + n + 1}$ is a positive integer.
4. Find all positive integers x, y such that $2^x + 3^y$ is a perfect square.
5. Find all nonnegative integers t, x, y, z such that $5^t + 3^x 4^y = z^2$.
6. Find all positive integers n such that there exists a prime number p such that $p^n - (p-1)^n$ is a power of 3.
7. Find the largest positive integer n that divides $p^6 - 1$ for all primes $p > 7$.
8. Find all integers a, b, c such that $\dfrac{(a-b)(b-c)(c-a)}{2} + 2$ is a power of 2016.
9. Find the maximum number of natural numbers $x_1, x_2, \cdots, x_m$ satisfying the conditions:

   (a) No $x_i - x_j, 1 \leq i < j \leq m$ is divisible by 11, and

   (b) The sum $x_2 x_3 \cdots x_m + x_1 x_3 \cdots x_m + \cdots + x_1 x_2 \cdots x_{m-1}$ is divisible by 11.

Classwork

20 July:

5.1 Exponent Q 6,10,11,13

Primitive Roots Examples 21, 22, 42, 47

Problems in Quadratic Congruencies (IMO Math) Problems 10, 14

23 July:

IMO Shortlist 03N1, 06N1-2, 09N1, 15N1